

Climate-Safe Housing Candidate Grid Model

What This Adds

This model converts Maine town screening results into statewide 5 km planning grid units. Each grid cell inherits town-level housing need, social vulnerability, and infrastructure proxy scores, then adds exposure signals from loaded hazard zones, environmental constraints, and nearby climate/infrastructure assets.

The result is a first grid-level suitability screen. It is more spatially explicit than town rankings, but it is not a parcel suitability determination. This build includes bounded FEMA NFHL polygons, the queryable Maine conserved-land service, and climate/infrastructure assets. Complete FEMA and NWI coverage, Maine Geological Survey inundation, DEM/slope, zoning, water/wastewater, and parcels are still required before final build/no-build decisions.

Model Formula

Component	Weight	Meaning
Housing need	30%	Prioritizes places where housing pressure is visible.
Infrastructure access proxy	30%	Rewards locations in towns with stronger service/access indicators and nearby road structures.
Low flood exposure	20%	Penalizes loaded flood/SLR hazard and exposed-asset signals.
Low environmental constraint	10%	Penalizes loaded wetland/conservation/environmental constraint intersections.
Lower social vulnerability	10%	Avoids treating highly vulnerable communities as simple growth targets without safeguards.

Output Snapshot

KPI	Value
Candidate grid units in PostGIS	4,800
Grid units exported to GeoJSON	4,800
Strong candidate units	173
Moderate candidate units	3,877
Avoid or detailed hazard review units	42
Resilience-before-growth units	0
Loaded hazard polygons	1,000
Loaded environmental constraints	156
Loaded infrastructure/exposed assets	1,188

Loaded Spatial Evidence

Role	Source	Features in PostGIS
environmental constraint	Maine GeoLibrary Conserved Lands	156
hazard polygon	FEMA National Flood Hazard Layer	1,000
infrastructure/exposed asset	Maine DEP Sea Level Rise Study	688
infrastructure/exposed asset	MaineDOT OpenData - Bridges	250
infrastructure/exposed asset	MaineDOT OpenData - Cross Culverts	250

Suitability Tier Counts

Tier	Grid Units
moderate candidate for parcel screening	3,877
monitor after full overlays	708
strong candidate for parcel screening	173
avoid or detailed hazard review	42

Highest Scoring Candidate Grid Units

Grid Unit	Town	County	Score	Tier	Flood	Env	Infra	Housing	Vulnerability
grid5km_2300560545_2478	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2372	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2587	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2481	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2479	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2693	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2480	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2586	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2053	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2371	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2373	Portland	Cumberland	81.20	strong candidate for parcel screening	0.00	0.00	84.36	67.34	43.12

Grid Unit	Town	County	Score	Tier	Flood	Env	Infra	Housing	Vulnerability
grid5km_2300560545_2374	Portland	Cumberland	78.60	strong candidate for parcel screening	16.00	0.00	86.36	67.34	43.12
grid5km_2300138740_2168	Lewiston	Androscoggin	75.32	strong candidate for parcel screening	0.00	0.00	66.55	70.39	57.61
grid5km_2300138740_2061	Lewiston	Androscoggin	74.96	strong candidate for parcel screening	4.78	0.00	68.55	70.39	57.61
grid5km_2300138740_1955	Lewiston	Androscoggin	74.16	strong candidate for parcel screening	8.78	0.00	68.55	70.39	57.61
grid5km_2300330690_3741	Hamlin	Aroostook	73.94	strong candidate for parcel screening	0.00	0.00	63.96	70.50	64.00
grid5km_2300138740_2167	Lewiston	Androscoggin	73.40	strong candidate for parcel screening	3.62	0.00	62.55	70.39	57.61
grid5km_2300138740_2060	Lewiston	Androscoggin	73.33	strong candidate for parcel screening	6.97	0.00	64.55	70.39	57.61
grid5km_2300138740_2062	Lewiston	Androscoggin	73.29	strong candidate for parcel screening	10.14	0.00	66.55	70.39	57.61
grid5km_2301902795_3796	Bangor	Penobscot	72.97	strong candidate for parcel screening	0.00	0.00	61.89	66.36	55.07

Grid Units Requiring Avoidance Or Detailed Hazard Review

Grid Unit	Town	County	Score	Tier	Flood	Env	Infra	Housing	Vulnerability
grid5km_2300560545_2266	Portland	Cumberland	65.80	avoid or detailed hazard review	80.00	0.00	86.36	67.34	43.12
grid5km_2300560545_2265	Portland	Cumberland	65.20	avoid or detailed hazard review	80.00	0.00	84.36	67.34	43.12
grid5km_2300560545_2051	Portland	Cumberland	63.20	avoid or detailed hazard review	96.00	0.00	88.36	67.34	43.12
grid5km_2300560545_2158	Portland	Cumberland	62.40	avoid or detailed hazard review	100.00	0.00	88.36	67.34	43.12
grid5km_2300560545_2159	Portland	Cumberland	62.40	avoid or detailed hazard review	100.00	0.00	88.36	67.34	43.12
grid5km_2300560545_2052	Portland	Cumberland	62.40	avoid or detailed hazard review	100.00	0.00	88.36	67.34	43.12
grid5km_2300571990_2158	South Portland	Cumberland	54.13	avoid or detailed hazard review	100.00	0.00	69.01	56.97	36.60
grid5km_2301902795_3901	Bangor	Penobscot	53.77	avoid or detailed hazard review	96.00	0.00	61.89	66.36	55.07
grid5km_2300571990_2265	South Portland	Cumberland	53.73	avoid or detailed hazard review	96.00	0.00	65.01	56.97	36.60
grid5km_2300571990_2051	South Portland	Cumberland	53.53	avoid or detailed hazard review	100.00	0.00	67.01	56.97	36.60

Grid Unit	Town	County	Score	Tier	Flood	Env	Infra	Housing	Vulnerability
grid5km_2301506120_3021	Boothbay Harbor	Lincoln	53.27	avoid or detailed hazard review	80.00	0.00	44.29	73.17	59.66
grid5km_2301363660_3780	Rockport	Knox	53.20	avoid or detailed hazard review	88.00	0.00	53.33	59.63	30.88
grid5km_2301363590_3778	Rockland	Knox	50.82	avoid or detailed hazard review	100.00	0.00	58.49	59.93	47.11
grid5km_2301186515_2608	Winslow	Kennebec	50.55	avoid or detailed hazard review	80.00	0.00	47.58	54.58	41.03
grid5km_2300582105_1945	Westbrook	Cumberland	50.38	avoid or detailed hazard review	100.00	0.00	61.29	51.15	33.55
grid5km_2300582105_2052	Westbrook	Cumberland	50.38	avoid or detailed hazard review	100.00	0.00	61.29	51.15	33.55
grid5km_2303164675_1940	Saco	York	50.37	avoid or detailed hazard review	100.00	0.00	58.81	53.19	32.27
grid5km_2300972865_4962	Southwest Harbor	Hancock	50.29	avoid or detailed hazard review	100.00	0.00	45.18	70.39	43.84
grid5km_2301102100_2496	Augusta	Kennebec	50.18	avoid or detailed hazard review	100.00	0.00	54.05	65.03	55.44
grid5km_2301309725_3780	Camden	Knox	49.49	avoid or detailed hazard review	96.00	0.00	53.28	56.19	41.54

Resilience-Before-Growth Grid Units

Grid Unit	Town	County	Score	Tier	Flood	Env	Infra	Housing	Vulnerability
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County Portfolio Summary

County	Grid Units	Avg Score	Strong Candidates	Hazard Review
Cumberland	297	65.74	33	11
York	306	64.32	26	3
Hancock	421	64.32	24	4
Androscoggin	127	63.51	19	1
Aroostook	628	61.95	19	1
Penobscot	526	61.96	17	2
Knox	227	63.97	15	9
Kennebec	236	64.11	11	2
Franklin	219	61.57	9	0
Sagadahoc	94	63.33	0	2
Waldo	228	63.01	0	0

County	Grid Units	Avg Score	Strong Candidates	Hazard Review
Washington	443	62.26	0	3
Oxford	357	62.09	0	0
Lincoln	179	62.04	0	4
Somerset	325	60.62	0	0
Piscataquis	187	60.61	0	0

How To Use This Output

- Use strong and moderate candidate grid units to choose where parcel screening should start.
- Use avoid/review units as a warning layer, not as a final legal restriction.
- Use resilience-before-growth units to prioritize infrastructure, anti-displacement, emergency access, and recovery-capacity investment.
- Use the GeoJSON in QGIS or the web app as a planning overlay with the existing town-screening outputs.

Next Precision Step

Replace the 5 km grid with parcel or 250 m grid cells in pilot towns after full hazard, wetlands, conservation, DEM/slope, zoning, roads, bridges, culverts, water/wastewater, and parcel layers are loaded.